PhD Formulas & Tricks — Made Easy Part 1

VCE Mathematical Methods Units 3 & 4 · Functions & graphs · Trigonometry · Composites & transformations

This is the short, friendly version of the big formula sheet. Every rule comes with a picture, a table or a worked example. Look for the coloured boxes:

TRICK — saves time | TRAP — where marks are lost | CHECK — catch your own mistakes | EXAMPLE — see it worked out

1 Functions and graphs

Interval notation — square bracket or round bracket?

Write	Means	On a graph	Example
`[a, b]`	both ends included	● filled dots at both ends	`[1, 4]` → $1 \le x \le 4$
`(a, b)`	both ends left out	○ open circles at both ends	`(1, 4)` → $1 < x < 4$
`[a, b)`	only a included	● at a, ○ at b	`[1, 4)` → $1 \le x < 4$
`R \ {2}`	every real number except 2	hole / asymptote at x = 2	`(−∞, 2) ∪ (2, ∞)`

∞ always gets a round bracket — you can never "reach" infinity.

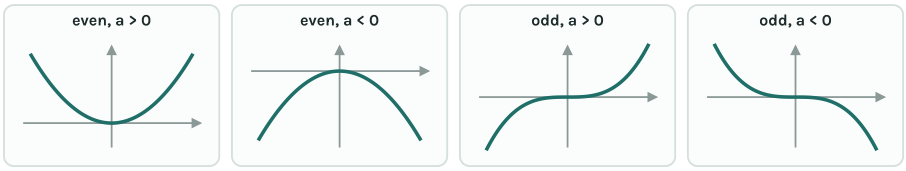
Maximal (largest possible) domain

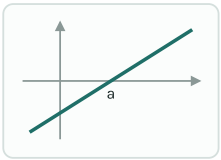
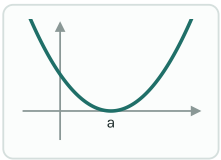
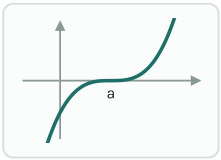
If the rule has…	You need…	Example	Domain
a square root √(g(x))	`g(x) ≥ 0`	`√(x − 3)`	`[3, ∞)`
a fraction 1 / g(x)	`g(x) ≠ 0`	`1 / (x + 5)`	`R \ {−5}`
a log $\log_e(g(x))$	`g(x) > 0`	$\log_e(2 - x)$	`(−∞, 2)`
more than one of these	all of them at once	see the example below	

EXAMPLE $y = \sqrt{(x + 2)} + 1/(x - 1)$. Root: $x \ge -2$. Fraction: $x \ne 1$. Together: **$[-2, 1) \cup (1, \infty)$**.

Polynomials — shape from the degree

A polynomial of degree **n** has **at most n** x-intercepts and **at most n − 1** turning points. Only the **leading term** decides where the arms go:



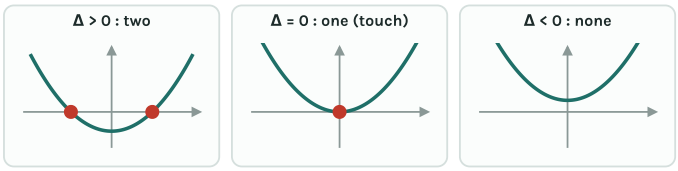
Factor	Picture near x = a	What the graph does	Sign changes?
$(x - a)$		**cuts** straight through	yes
$(x - a)^2$		**touches** and turns back (turning point)	no
$(x - a)^3$		**flattens** then goes through (stationary point of inflection)	yes

Rule of thumb: **even** power → touch, **odd** power → cross.

EXAMPLE $y = (x - 1)^2(x + 3)$: touches the x-axis at x = 1, cuts at x = −3. It is a cubic but has only **2 distinct** zeros.

Quadratics $y = ax^2 + bx + c$

What you want	Formula	Quick note
x of the turning point	$x = -b / (2a)$	then substitute to get y
turning point form	$y = a(x - h)^2 + k \rightarrow TP\ (h, k)$	watch the minus: $y = 2(x + 3)^2 - 1$ has TP (−3, −1)
quadratic formula	$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$	leave surds exact
axis of symmetry	$x = (p + q) / 2$	halfway between the x-intercepts p and q
discriminant	$\Delta = b^2 - 4ac$	tells you **how many** solutions — see pictures



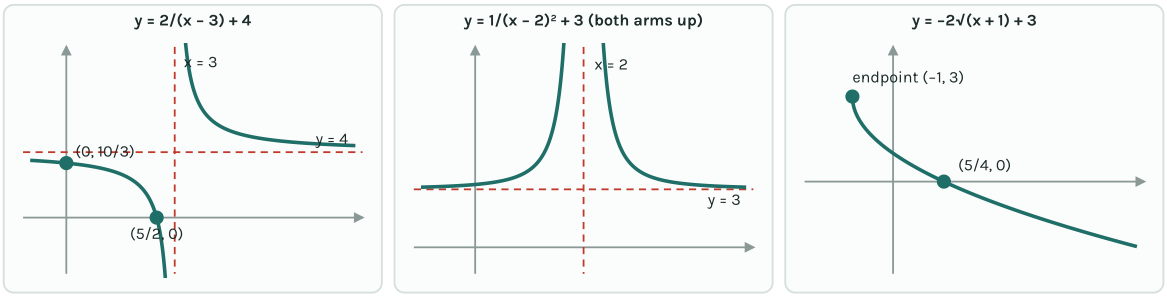
TRICK

Use the discriminant — do not solve

"For what k is the line a tangent?" → set the rules equal, collect into $ax^2 + bx + c = 0$, then put **$\Delta = 0$**. ($\Delta > 0$: two intersections, $\Delta < 0$: none.) No need to actually solve the quadratic.

Hyperbola, truncus and square-root graphs

Graph	Rule	Asymptotes / endpoint	Domain	Range
Hyperbola	$y = a/(x - h) + k$	x = h, y = k	$R \setminus \{h\}$	$R \setminus \{k\}$
Truncus	$y = a/(x - h)^2 + k$	x = h, y = k	$R \setminus \{h\}$	(k, ∞) if $a > 0$
Square root	$y = a\sqrt{(x - h)} + k$	endpoint (h, k)	$[h, \infty)$	$[k, \infty)$ if $a > 0$ $(-\infty, k]$ if $a < 0$



EXAMPLE $y = 2/(x - 3) + 4$ x-intercept: $2/(x - 3) = -4 \rightarrow x - 3 = -\frac{1}{2} \rightarrow x = 5/2$. y-intercept: $2/(-3) + 4 = 10/3$.

TRICK

Split the numerator instead of long division

Make the top contain the bottom: $(3x + 1)/(x - 2) = (3(x - 2) + 7)/(x - 2) = 3 + 7/(x - 2)$. Now you can read the asymptotes straight off: **x = 2** and **y = 3**.

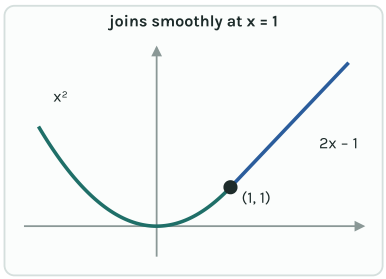
CHECK

Endpoint or asymptote? Pick the right bracket

A square-root endpoint **is** reached → square bracket: range of $y = -2\sqrt{(x + 1)} + 3$ is $(-\infty, 3]$. An asymptote is **never** reached → round bracket / "\": range of the truncus above is $(3, \infty)$.

Hybrid (piecewise) functions

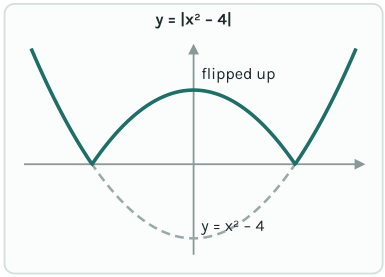
To be…	Check at the join x = a
continuous (no gap)	left value = right value = f(a)
differentiable (smooth, no corner)	continuous **and** left gradient = right gradient



EXAMPLE $f(x) = x^2$ for $x \le 1$, $2x - 1$ for $x > 1$.
Values: $1^2 = 1$ and $2(1) - 1 = 1$ ✓ Gradients: $2x = 2$ and 2 ✓
→ smooth join.
If it were x^2 and x: values match ($1 = 1$) but gradients $2 \neq 1$
→ a **corner**, not differentiable.

Modulus |x| and odd / even functions

Type	Rule	Example
equation	$\|x - a\| = b \to x = a \pm b$	$\|x - 2\| = 5 \to x = 7$ or -3
"less than"	$\|x\| < a \Leftrightarrow -a < x < a$	$\|x\| < 3 \to (-3, 3)$
"greater than"	$\|x\| > a \Leftrightarrow x < -a$ or $x > a$	two separate pieces
graph of $\|f(x)\|$	flip everything **below** the x-axis up	
graph of $f(\|x\|)$	keep $x \ge 0$, mirror it in the y-axis	



Even function	Odd function
$f(-x) = f(x)$ — mirror image in the y-axis. e.g. x^2, cos x, $\|x\|$	$f(-x) = -f(x)$ — same after a half-turn about the origin. e.g. x^3, sin x, $1/x$

TRAP

Double factor touches, triple factor flattens

At $(x - a)^2$ the curve turns back. At $(x - a)^3$ it flattens and **keeps going**. Students often draw a turning point at a triple factor — don't.

CHECK

Count features against the degree

A degree-n polynomial can't have more than n x-intercepts or more than n − 1 turning points. A quartic drawn with 4 turning points is wrong on sight.

Try it yourself — Section 1

1. Find the maximal domain of $y = \sqrt{(4 - x)} + 1/(x + 1)$.
2. Find the asymptotes of $y = (2x + 5)/(x + 1)$. (Hint: split the numerator.)
3. Describe what $y = (x + 2)^2(x - 1)^3$ does at $x = -2$ and at $x = 1$.

2 Trigonometric functions

Radians and exact values

`π rad = 180°` · degrees → radians: × π/180 · radians → degrees: × 180/π

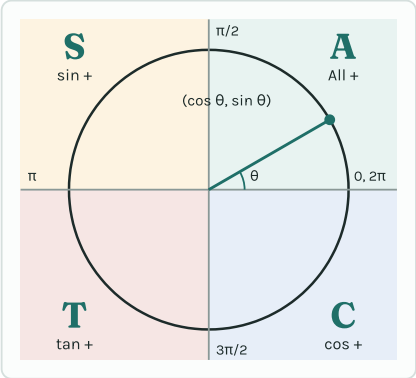
θ	0	π/6 (30°)	π/4 (45°)	π/3 (60°)	π/2 (90°)
sin θ	0	1/2	√2/2	√3/2	1
cos θ	1	√3/2	√2/2	1/2	0
tan θ	0	√3/3 (= 1/√3)	1	√3	undefined

TRICK

Remember the sin row in 5 seconds

Write `√0/2, √1/2, √2/2, √3/2, √4/2` → 0, ½, √2/2, √3/2, 1. The cos row is the **same numbers backwards**. tan = sin ÷ cos.

The unit circle and the quadrants (ASTC)



"All Students Take Calculus" — which ratio is positive

Quadrant	Angle in terms of base angle θ	Positive
1 (0 to π/2)	`θ`	all
2 (π/2 to π)	`π − θ`	sin only
3 (π to 3π/2)	`π + θ`	tan only
4 (3π/2 to 2π)	`2π − θ`	cos only

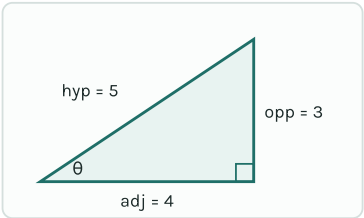
EXAMPLE sin(7π/6): 7π/6 = π + π/6 → quadrant 3, base angle π/6. sin is negative there, so **sin(7π/6) = −1/2**.

Also useful: `sin(−θ) = −sin θ`, `cos(−θ) = cos θ`, `sin(π/2 − θ) = cos θ`, `sin²θ + cos²θ = 1`, `tan θ = sin θ / cos θ`.

Finding one ratio from another

EXAMPLE sin θ = 3/5 and θ ∈ (π/2, π). Find cos θ and tan θ.

① Draw the triangle: opp 3, hyp 5 → adj 4 (3-4-5).
② Quadrant 2: cos is negative → **cos θ = −4/5**.
③ **tan θ = (3/5) ÷ (−4/5) = −3/4**.



the triangle gives the size, the quadrant gives the sign

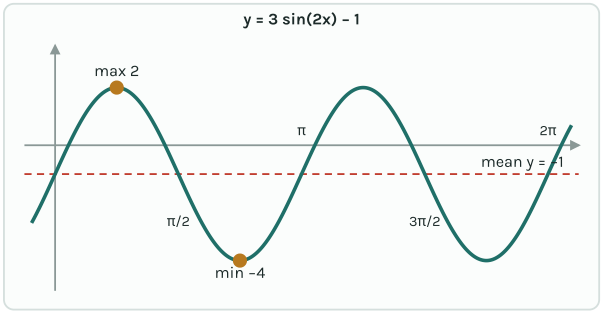
TRAP

The identity gives the size, the quadrant gives the sign

cos²θ = 1 − 9/25 = 16/25 gives cos θ = ±4/5. Only the given domain tells you which sign. Writing +4/5 loses the mark even though the arithmetic is right.

Graphs of sine and cosine: y = a sin(n(x + b)) + c

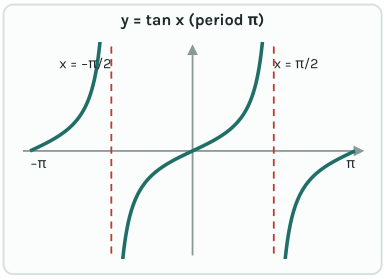
Feature	Formula	For y = 3 sin(2x) − 1
amplitude	`|a|`	3
period	`2π / |n|`	2π/2 = π
mean (centre) line	`y = c`	y = −1
range	`[c − |a|, c + |a|]`	[−4, 2]
cycles in [0, 2π]	`2π ÷ period`	2 cycles



Key points are a **quarter-period apart**: sine starts at the mean line going up; cosine starts at a max (if a > 0).

The tangent graph y = a tan(n(x + b)) + c

Feature	Rule
period	`π / |n|`
asymptotes of tan x	`x = π/2 + kπ, k ∈ Z`
x-intercepts of tan x	`x = kπ, k ∈ Z`
amplitude	**none** — range is R



EXAMPLE y = tan(2x): period π/2. Asymptotes: 2x = π/2 + kπ → x = π/4 + kπ/2.

Solving a trig equation on a restricted domain — 4 steps

Example: solve `2 sin(2x) + 1 = 0` for `x ∈ [0, 2π]`.

1 Isolate the trig ratio
2 sin(2x) + 1 = 0 → sin(2x) = −½

→

2 Change the domain
x ∈ [0, 2π] → 2x ∈ [0, 4π]

→

3 Base angle → quadrants → add periods
base π/6; sin − in Q3, Q4: 7π/6, 11π/6, then +2π: 19π/6, 23π/6

→

4 Undo (÷ 2)
x = 7π/12, 11π/12, 19π/12, 23π/12

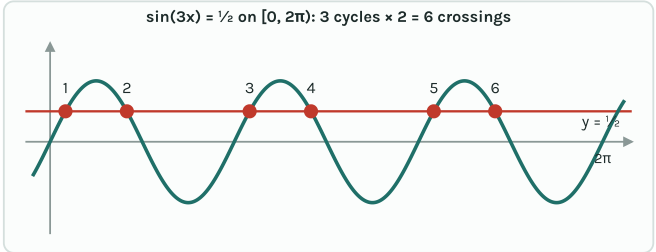
TRICK

Change the domain, not the answers

Convert the domain first (2x ∈ [0, 4π]), find **every** answer for 2x in that bigger interval, then divide by 2 at the very end. Solving for x first and adding periods later is where solutions go missing.

How many solutions should I expect?

Equation (over one interval of length 2π)	Number of solutions	Example on [0, 2π)
`sin(nx) = k or cos(nx) = k, with |k| < 1`	**2n**	`sin(3x) = ½ → 6 solutions`
`sin(nx) = ±1 or cos(nx) = ±1`	**n** (one per cycle)	`sin(2x) = 1 → 2 solutions`
`tan(nx) = k (any k)`	**n per π**, so 2n per 2π	`tan x = −√3 → 2 solutions`
`sin or cos = k with |k| > 1`	**0**	`sin x = 2 → no solution`



EXAMPLE sin(3x) = ½ on [0, 2π): 3x ∈ [0, 6π). Base angle π/6, sin + in Q1, Q2:
3x = π/6, 5π/6, 13π/6, 17π/6, 25π/6, 29π/6 → **x = π/18, 5π/18, 13π/18, 17π/18, 25π/18, 29π/18** — six answers, matching the graph. **If you have fewer, you dropped some.**

CHECK

Range check kills impossible equations

y = 3 sin(2x) − 1 only takes values in [−4, 2]. So `3 sin(2x) − 1 = 3` has **no solution** — check this before you start solving.

General solutions (n ∈ Z)

Equation	General solution	Example
`sin x = a`	`x = 2nπ + sin⁻¹(a) or x = (2n + 1)π − sin⁻¹(a)`	`sin x = ½ → x = 2nπ + π/6 or (2n+1)π − π/6`
`cos x = a`	`x = 2nπ ± cos⁻¹(a)`	`cos x = ½ → x = 2nπ ± π/3`
`tan x = a`	`x = nπ + tan⁻¹(a)`	`tan x = 1 → x = nπ + π/4`

CHECK

Put n = 0, 1, −1 back in

Always write **n ∈ Z**. Then substitute n = 0, 1, −1 — the values that land in the domain should match your restricted-domain answers.

TRAP

Factorise before you read the translation

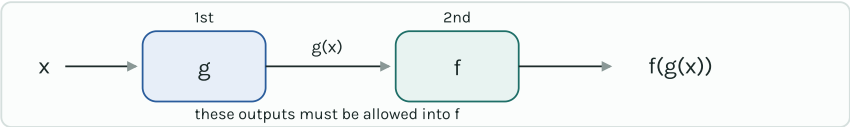
`sin(2x − π/3)` is **not** a shift of π/3. Write it as `sin(2(x − π/6))`: the shift is **π/6 to the right**. Any time n ≠ 1, take n out of the bracket first.

Try it yourself — Section 2

1. How many solutions does `cos(2x) = 0.3` have for x ∈ [0, 2π]?
2. Solve `cos x = −½` for x ∈ [0, 2π].
3. State the amplitude, period and range of `y = −2 cos(3x) + 5`.

3 Composite functions and transformations

Composite functions f(g(x)) — think of two machines



Fact	Rule	What it means
Order	`f(g(x))`	do **g first**, then f (also written f ∘ g)
Exists when	`ran g ⊆ dom f`	everything g puts out must be allowed into f
Domain	`dom f(g(x)) = dom g`	after any restriction you had to make
Not symmetric	`f(g(x)) ≠ g(f(x))`	one can exist while the other does not

EXAMPLE — IT FAILS, THEN FIX IT f(x) = √x, g(x) = x − 3 (dom R).
ran g = R, dom f = [0, ∞). R ⊄ [0, ∞) → f(g(x)) **does not exist**.
Fix: restrict dom g to [3, ∞). Then ran g = [0, ∞) ✓ and f(g(x)) = √(x − 3).

EXAMPLE — RANGE OF A COMPOSITE f(x) = x², g(x) = x − 1, dom g = [0, 4].
Step 1: ran g = [−1, 3].
Step 2: square that set → **[0, 9]**, **not** [1, 9], because x = 1 gives 0, the minimum.

TRAP

It is ran g ⊆ dom f, not dom g ⊆ dom f

The test is on the **outputs** of the inside function. Write down ran g, write down dom f, then compare those two sets.

The transformation rule y = a f(b(x + c)) + d

Letter	Effect	Exam wording	Example from y = f(x)
a	multiplies every **y** by a (a < 0 also flips)	dilation of factor |a| from the x-axis	`y = 3f(x): 3 × taller`
b	**divides** every **x** by b (b < 0 also flips)	dilation of factor 1/|b| from the y-axis	`y = f(2x): squashed, factor ½`
c	moves left by c	translation of c units in the negative x direction	`y = f(x + 3): 3 left`
d	moves up by d	translation of d units in the positive y direction	`y = f(x) + 2: 2 up`

Image of a point: `(p, q) → (p/b − c, aq + d)`. **Order:** dilations and reflections first, translations last.

Example: y = 2f(3(x − 1)) + 5 sends the point (2, 4) to …		
Coordinate	**Working**	**Result**
x	divide by 3, then add 1 → 2/3 + 1	**5/3**
y	multiply by 2, then add 5 → 2 × 4 + 5	**13**
Check with f(x) = x²: 2(3(5/3 − 1))² + 5 = 2(2)² + 5		= 13 ✓

TRAP

Dilation factors from the y-axis go upside down

f(2x) is a dilation of factor ½ from the y-axis, not 2 — the graph is squashed towards the y-axis. Vertical ones read straight off: 2f(x) is factor 2 from the x-axis.

TRICK

Asymptotes and endpoints move too

Apply the point rule to x = h and y = k as well. For y = 1/x → y = 2/(x − 3) + 4, the asymptotes x = 0, y = 0 move to **x = 3, y = 4**. Fastest way to sketch any transformed graph.

Try it yourself — Section 3

1. f(x) = √x and g(x) = x + 2 with dom g = R. Does f(g(x)) exist? If not, give the largest domain of g that fixes it.
2. Find the image of the point (1, 3) under `y = −f(2(x + 1)) + 4`.
3. Describe the transformations from y = sin x to `y = sin(2x − π/3)`.

Answers — Try it yourself

Section 1

1. x ≤ 4 and x ≠ −1 → `(−∞, −1) ∪ (−1, 4]`

2. (2x + 5)/(x + 1) = 2 + 3/(x + 1) → **x = −1, y = 2**

3. x = −2: touches (turning point). x = 1: flattens and passes through (stationary point of inflection).

Section 2

1. 2x ∈ [0, 4π] → 2 cycles × 2 = **4** solutions

2. Base angle π/3, cos − in Q2, Q3 → **x = 2π/3, 4π/3**

3. Amplitude 2, period 2π/3, range [3, 7]

Section 3

1. ran g = R ⊄ [0, ∞), so no. Restrict to **dom g = [−2, ∞)**.

2. x: 1 ÷ 2 − 1 = −½; y: −3 + 4 = 1 → **(−½, 1)**

3. sin(2(x − π/6)): dilation of factor ½ from the y-axis, then translation of π/6 units in the positive direction of the x-axis.